A SHORT COMBINATORIAL PROOF OF SEYMOUR'S SECOND NEIGHBOURHOOD CONJECTURE

QIHUAN XIE

Abstract. Seymour's second neighbourhood conjecture asserts that every nonempty finite oriented graph has a vertex whose second out-neighbourhood is at least as large as its first. A recent proof of the conjecture reduces it, through an extremal argument, to a deletion lemma for conflicting pairs of binary relations, which is proved there by a generic-rank argument. We give a combinatorial proof of this lemma, based on Hall's theorem and on greedy bases of transversal matroids taken along a single linear order. This yields a self-contained combinatorial proof of the conjecture. We also record a vertex-weighted version and an application to directed triangles.

1. Introduction

An *oriented graph* is a finite directed graph without loops or digons (pairs of opposite arcs). Given a vertex v in an oriented graph D, we denote by $N_1^+(v) = \{u \in V(D) : v \to u\}$ its first out-neighbourhood, and by $N_2^+(v)$ the set of vertices at directed distance precisely two from v. More generally, for any subset $S \subseteq V(D)$, we write

$$F(S) = \{y \in V(D) : \ x \to y \text{ for some } x \in S\} \qquad \text{and} \qquad F^2(S) = F(F(S)).$$

Note that $F(S)$ is permitted to intersect S. Because an oriented graph contains no directed closed walks of length two, the second out-neighbourhood of a single vertex v is $N_2^+(v) = F^2(\{v\})\setminus F(\{v\})$.

In 1990, Paul Seymour conjectured that every nonempty oriented graph possesses a vertex whose second out-neighbourhood is at least as large as its first.

Theorem 1.1 (Seymour's second neighbourhood conjecture)**.** *Every nonempty oriented graph contains a vertex v satisfying $|N_1^+(v)| \le |N_2^+(v)|$.*

First recorded in print by Dean and Latka [4], who posed the special case of tournaments, Seymour's conjecture has been studied extensively. The tournament case was settled by Fisher [8] by a probabilistic argument based on Farkas' lemma, and subsequently by Havet and Thomassé [11] using the combinatorial method of median orders. Building on these techniques, Fidler and Yuster [7] and Ghazal [9] established the conjecture for several dense families of oriented graphs, such as tournaments missing a star or a subtournament, while Kaneko and Locke [14] verified it for oriented graphs of minimum outdegree at most six. For arbitrary oriented graphs on n vertices, Chen, Shen, and Yuster [3] proved the existence of a vertex v with $|N_2^+(v)| \ge \gamma|N_1^+(v)|$ for $\gamma \approx 0.657$, a constant recently improved to $\gamma \approx 0.7155$ by Huang and Peng [13]. Furthermore, Espuny Díaz, Girão, Granet, and Kronenberg [6] showed that for any fixed $p < 1/2$, almost every orientation of the binomial random graph $G(n,p)$ satisfies the conjecture. Recently, a proof of the full conjecture was given in [15].

The proof in [15] is combinatorial except for one step, a deletion lemma whose proof is algebraic. The purpose of this paper is to give a combinatorial proof of that lemma, and thereby a self-contained combinatorial proof of Theorem 1.1.

Date: November 7, 2026.

2020 *Mathematics Subject Classification.* Primary 05C20; Secondary 05B35, 05C70, 05D15.

Key words and phrases. second neighbourhood conjecture, oriented graphs, Caccetta–Häggkvist conjecture, Hall's theorem, transversal matroids.

Call a vertex v *good* if $|N_1^+(v)| \le |N_2^+(v)|$, and *bad* otherwise. As in [15], which builds on Brantner, Brockman, Kay, and Snively [1] and Seacrest [17], the argument has three steps.

(1) Using the vertex- and arc-minimality of a counterexample D, one shows that every nonempty proper subset $S \subsetneq V(D)$ satisfies $|F^2(S) \setminus F(S)| < |F(S) \setminus S|$ (Lemma 2.1).
(2) Replacing each vertex of D by a transitive tournament of order $m > n$ multiplies this deficit by m, at the cost of an error of at most one per fibre. The result is an oriented graph G with positive minimum indegree in which $|U| + |F^2(U)| < 2|F(U)|$ for every nonempty proper subset $U \subsetneq V(G)$ (Lemma 2.2).
(3) Finally, one shows that no such G exists:

Theorem 1.2 (Set version)**.** *Every nonempty oriented graph with minimum indegree at least one contains a nonempty proper vertex subset U satisfying $|U| + |F^2(U)| \ge 2|F(U)|$.*

Theorem 1.2 is proved in Section 4 by the extremal argument of [15] on $X \times X$, where $X = V(G)$. One considers pairs (P, Q) of relations on X that are *compatible*: applying F to every column of P and to every row of Q gives disjoint subsets of $X \times X$. The diagonal pair is compatible exactly because G has no digons, so compatibility is the digon-free condition lifted to pairs of relations. For a compatible pair maximizing a suitable potential, the concavity inequality, applied to every column of P and every row of Q, produces two families R and C of larger total size. These need not be compatible, and the cost of removing the resulting conflicts is controlled by the following lemma.

Theorem 1.3 (Deletion lemma, informal version of Theorem 3.1)**.** *Let R and C be two families of pairs, and call $r \in R$ and $c \in C$* conflicting *when they close up a square as in Figure 1(a); each conflict has a Z-point and an H-point. Then one can delete at most $|H| + q$ members of $R \cup C$ so that no conflicts remain, where H is the set of H-points and q is the number of Z-points covered by no remaining member.*

In [15] this lemma is derived from a matching bound for an auxiliary bipartite graph on duplicated Z-points, which is proved using Edmonds' theorem on generic ranks [5], augmenting paths, and a kernel lemma from linear algebra. Our proof (Section 5) has two cases. If the H-points cannot be matched into one of the families, the defect form of Hall's theorem gives a small set of members whose removal destroys many H-points, and we induct. Otherwise we delete *everything*, and must show that $|R| + |C| \le |Z| + |H|$. In the algebraic approach this follows from Sylvester's rank inequality for a factorization $NL = AB$ of a "conflict matrix" through Z and through H. We replace it by a matching argument (Lemma 5.4): for each row we take a greedy basis of a transversal matroid, always with respect to one fixed linear order. The point is that an element that is greedy for a single row remains greedy when several rows are pooled, so for each p the greedy elements of all rows met by p can be matched simultaneously, to distinct H-points.

Section 2 contains the two reductions, Section 3 the general setting and the precise statement of the deletion lemma, Section 4 the extremal argument, and Section 5 the proof of the deletion lemma. Section 6 records a vertex-weighted version of Theorem 1.1 and the known consequence for directed triangles, which settles the minimum-semidegree version of the triangle case of the Caccetta–Häggkvist conjecture.

2. Reduction to the set version

We first show that in a vertex- and arc-minimal counterexample to Theorem 1.1, every nonempty proper vertex subset satisfies the deficit inequality (1) below. The arc-deletion argument below is a streamlined adaptation of [17, Lemma 4].

Lemma 2.1 (Deficit lemma)**.** *Let D be a counterexample to Theorem 1.1 having the minimum number of vertices and, subject to this, the minimum number of arcs. Then $F(S) \not\subseteq S$ for every*

nonempty proper subset $S \subsetneq V(D)$; in particular, every vertex of D has positive indegree. Moreover,

$$|F^2(S) \setminus F(S)| < |F(S) \setminus S| \qquad \text{for every nonempty proper } S \subsetneq V(D). \tag{1}$$

Proof. Clearly $|V(D)| \geq 2$, since a single vertex is good. If $F(S) \subseteq S$ for some nonempty proper S, then every $v \in S$ has the same first and second out-neighbourhoods in $D[S]$ as in D, so $D[S]$ is a smaller counterexample. Taking $S = V(D) \setminus \{y\}$ shows that every vertex y has an in-neighbour.

Fix a nonempty proper subset $S \subsetneq V(D)$, and partition $F(S)$ into its internal part $I = F(S) \cap S$ and its external boundary $E = F(S) \setminus S$; by the preceding paragraph, $E \neq \varnothing$. For each subset $T \subseteq E$, we measure the expansion of T outside $I \cup T$ by setting $\gamma(T) = |F(T) \setminus (I \cup T)|$.

We claim that for every proper subset $T \subsetneq E$, there exists a nonempty subset $A \subseteq E \setminus T$ such that $\gamma(T \cup A) - \gamma(T) < |A|$. To see this, form the oriented graph D_T from D by deleting all arcs directed from S to $E \setminus T$. Because every vertex in $E \setminus T \subseteq F(S)$ has an in-neighbour in S, at least one arc is removed, so by the arc-minimality of D the graph D_T must contain a good vertex v. Note that deleting arcs cannot create new directed walks, so if v lost no out-neighbours in passing to D_T, its first out-neighbourhood would remain unchanged while its second out-neighbourhood could only shrink, leaving v bad. Thus v must lose at least one out-neighbour, which forces $v \in S$ and ensures that the set $A = F(\{v\}) \cap (E \setminus T)$ of lost out-neighbours is nonempty. Since $F(\{v\}) \subseteq F(S) = I \cup E$, all remaining out-neighbours of v in D_T lie in $I \cup T$.

Now consider any two-step walk $v \to u \to w$ in D_T. Necessarily $u \in I \cup T$. If $u \in I \subseteq S$, then $w \in F(S) \setminus (E \setminus T) = I \cup T$, whereas if $u \in T$, then $w \in F(T)$. Hence

$$F^2_{D_T}(\{v\}) \subseteq I \cup T \cup F(T). \tag{2}$$

Let $k_1 = |N_1^+(v)|$ and $k_2 = |N_2^+(v)|$ denote the neighbourhood sizes of v in D, so that $k_2 < k_1$. Any vertex w that belongs to the second out-neighbourhood of v in D_T but not in D must lie in $F^2_D(\{v\}) \setminus (\{v\} \cup N_2^+(v)) \subseteq N_1^+(v)$ while failing to be an out-neighbour of v in D_T; thus $w \in A$, and combining (2) with $A \cap (I \cup T) = \varnothing$ shows that $w \in A \cap F(T)$. Conversely, consider the set $\Lambda = F(A) \setminus (I \cup T \cup A \cup F(T))$. Every vertex $w \in \Lambda$ is reached from v in D via a two-step walk through A; moreover, $w \neq v$ (as D has no digons) and $w \notin N_1^+(v) \subseteq I \cup T \cup A$. Thus $\Lambda \subseteq N_2^+(v)$, whereas (2) guarantees that no vertex of Λ remains a second out-neighbour of v in D_T. Since v is good in D_T, comparing its first and second out-degrees in D_T yields

$$k_1 - |A| \;\leq\; k_2 - |\Lambda| + |A \cap F(T)|.$$

On the other hand, augmenting T by A removes from $F(T) \setminus (I \cup T)$ precisely the set $A \cap F(T)$ and adjoins precisely Λ. Therefore,

$$\gamma(T \cup A) - \gamma(T) = |\Lambda| - |A \cap F(T)| \leq |A| + k_2 - k_1 < |A|,$$

as claimed.

Starting from $T = \varnothing$ and applying this claim iteratively (always within the fixed graph D), we exhaust the finite set E in finitely many steps. Telescoping the resulting strict inequalities and using $\gamma(\varnothing) = 0$ gives $\gamma(E) < |E|$. Finally, since $F(I) \subseteq F(S) = I \cup E$, we have $F^2(S) \setminus F(S) = F(I \cup E) \setminus (I \cup E) = F(E) \setminus (I \cup E)$, which has size $\gamma(E) < |E| = |F(S) \setminus S|$. This establishes (1). □

Next we blow up D, as in [1, Theorem 4.3], replacing each vertex by a transitive tournament on $m > n$ vertices.

Lemma 2.2 (Blow-up lemma)**.** *Let D be an oriented graph on n vertices with positive minimum indegree satisfying (1). For any integer $m > n$, let G be the graph on $V(D) \times \{1, \ldots, m\}$ with arcs $(v, a) \to (w, b)$ whenever either $v \to w$ in D, or $v = w$ and $a < b$. Then G is an oriented graph with positive minimum indegree satisfying*

$$|U| + |F^2_G(U)| < 2|F_G(U)| \qquad \text{for every nonempty proper } U \subsetneq V(G). \tag{3}$$

Proof. It is immediate from the definition that G is loopless and digon-free, and if $u \to w$ in D, then $(u,1)$ serves as an in-neighbour for every (w,b) in G.

For a subset $\Theta \subseteq \{1,\dots,m\}$, write $f(\Theta) = \{b : \ b > a \text{ for some } a \in \Theta\}$, with $f(\varnothing) = \varnothing$. If $\Theta \neq \varnothing$ and $a = \min \Theta$, then $|f(\Theta)| = m - a$, $|f^2(\Theta)| = \max(m-a-1, 0)$ and $|\Theta| \leq m - a + 1$, so

$$|\Theta| + |f^2(\Theta)| - 2|f(\Theta)| \leq 1. \tag{4}$$

Now fix a nonempty proper subset $U \subsetneq V(G)$. For each $v \in V(D)$, let $\Theta_v = \{a : (v,a) \in U\}$ be the v-fibre of U, let $S = \{v \in V(D) : \Theta_v \neq \varnothing\}$ be the projection of U, and write $F = F_D$. Under F_G, the first image $F_G(U)$ fills the entire fibre $\{w\} \times \{1,\dots,m\}$ over every $w \in F(S)$, while within each fibre $v \in S \setminus F(S)$ it occupies precisely $\{v\} \times f(\Theta_v)$. Thus,

$$|F_G(U)| = m|F(S)| + \sum_{v \in S \setminus F(S)} |f(\Theta_v)|.$$

Similarly, a two-step walk in G starting in U ends in a fibre over $F^2(S)$ if it uses two inter-fibre arcs, in a fibre over $F(S)$ if it uses one inter-fibre arc and one internal arc, and in $\{v\} \times f^2(\Theta_v)$ for some $v \in S$ if it uses two internal arcs. Writing $F^{\leq 2}(S) = F(S) \cup F^2(S)$, we see that $F_G^2(U)$ contains at most $m|F^{\leq 2}(S)|$ vertices in fibres over $F^{\leq 2}(S)$, and its remaining vertices lie in $\{v\} \times f^2(\Theta_v)$ for $v \in S \setminus F^{\leq 2}(S) \subseteq S \setminus F(S)$. Consequently,

$$|F_G^2(U)| \leq m|F^{\leq 2}(S)| + \sum_{v \in S \setminus F(S)} |f^2(\Theta_v)|.$$

Bounding $|\Theta_v| \leq m$ for $v \in S \cap F(S)$, applying (4) to each $v \in S \setminus F(S)$, and substituting $|F^{\leq 2}(S)| = |F(S)| + |F^2(S) \setminus F(S)|$, we obtain

$$\begin{aligned} |U| + |F_G^2(U)| - 2|F_G(U)| &\leq m\big(|S \cap F(S)| + |F^{\leq 2}(S)| - 2|F(S)|\big) + |S \setminus F(S)| \\ &= m\big(|F^2(S) \setminus F(S)| - |F(S) \setminus S|\big) + |S \setminus F(S)|. \end{aligned}$$

If $S \neq V(D)$, then (1) ensures that the parenthesized difference is at most -1, so the right-hand side is at most $-m + |S \setminus F(S)| \leq -m + n < 0$. If instead $S = V(D)$, then since every vertex of D has an in-neighbour we have $F(S) = V(D)$, which forces $F_G(U) = F_G^2(U) = V(G)$ and gives $|U| + |F_G^2(U)| - 2|F_G(U)| = |U| - |V(G)| < 0$. □

3. Conflicts and the deletion lemma

We state the deletion lemma for general relations. An *instance* $\mathcal{I} = (X_1, X_2, X_3, X_4, \alpha, \beta, R, C)$ consists of four finite ground sets X_1, X_2, X_3, X_4, two binary relations $\alpha \subseteq X_1 \times X_2$ and $\beta \subseteq X_3 \times X_4$, and two families of ordered pairs $R \subseteq X_1 \times X_4$ and $C \subseteq X_2 \times X_3$. Throughout this section, we adhere to the index conventions $p \in X_1$, $i \in X_2$, $s \in X_3$, and $j \in X_4$, writing $p \to i$ for $(p,i) \in \alpha$ and $s \to j$ for $(s,j) \in \beta$. We refer to the elements of R and C as *members*.[1]

Two members $r = (p,j) \in R$ and $c = (i,s) \in C$ are said to *conflict* if $p \to i$ and $s \to j$. Each such conflict determines a *Z-point* $(i,j) \in X_2 \times X_4$ and an *H-point* $(p,s) \in X_1 \times X_3$; we denote by Z and H the sets of all Z-points and H-points arising from conflicts in $\mathcal{I}$. Furthermore, we say that $r = (p,j) \in R$ *covers* $z = (i,j) \in Z$ if $p \to i$, and that $c = (i,s) \in C$ *covers* $z = (i,j) \in Z$ if $s \to j$. Geometrically, a conflict may be visualized as a square whose vertical sides are the members $r \in R$ and $c \in C$, whose horizontal directed edges are $p \to i$ and $s \to j$, and whose diagonals are the Z-point (i,j) and the H-point (p,s) (see Figure 1(a)).

If $r = (p,j) \in R$ and $c = (i,s) \in C$ cover the same point $z = (i,j) \in Z$, then $p \to i$ and $s \to j$, so r and c conflict. Since every $z \in Z$ arises from at least one conflict, it follows that any member

[1]When the ground sets coincide, as in Section 4, R and C are still regarded as disjoint families, and likewise the two sides of each bipartite graph below are taken disjoint.

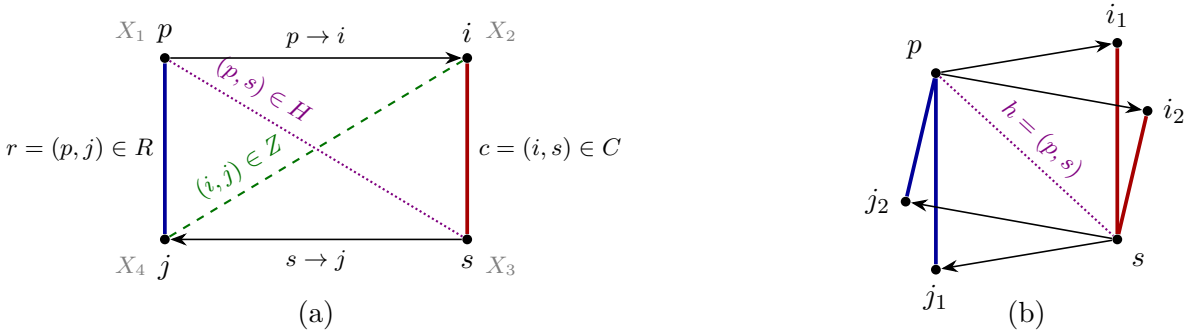


FIGURE 1. (a) A conflict between $r = (p, j) \in R$ and $c = (i, s) \in C$, covering the Z-point (i, j) via $p \to i$ and $s \to j$. The involution σ (defined below) acts as a half-turn of the square, exchanging R with C and α with β while transposing both diagonals. (b) A conflict block $R(h) \times C(h)$ over $h = (p, s)$: every member $(p, j_b) \in R(h)$ conflicts with every member $(i_a, s) \in C(h)$, producing distinct Z-points (i_a, j_b).

of R or C covering a point in Z must itself participate in a conflict. Moreover, if we define for each $h = (p, s) \in H$ the nonempty sets

$$R(h) = \{(p, j) \in R : \ s \to j\} \qquad \text{and} \qquad C(h) = \{(i, s) \in C : \ p \to i\},$$

then the conflicts having H-point h are precisely the pairs in the complete bipartite block $R(h) \times C(h)$ (see Figure 1(b)).

We associate with $\mathcal{I}$ four bipartite graphs, named after their two sides:

- Γ_{HR} (between H and R) and Γ_{HC} (between H and C), where $h \in H$ is adjacent to the members of $R(h)$ and $C(h)$, respectively;
- Γ_{RZ} (between R and Z) and Γ_{CZ} (between C and Z), where a member of R or C is adjacent to the points of Z that it covers.

Throughout, we write $\nu(\Gamma)$ for the maximum matching size in a bipartite graph Γ.

Theorem 3.1 (Deletion lemma)**.** *For every instance $\mathcal{I}$, there exist conflict-free subfamilies $R' \subseteq R$ and $C' \subseteq C$, retaining all members of R and C that participate in no conflicts, such that*

$$|R \setminus R'| + |C \setminus C'| \le \min\{\nu(\Gamma_{HR}), \nu(\Gamma_{HC})\} + q \le |H| + q,$$

where q is the number of points in Z covered by no member of R' or C'.

Theorem 3.1 is used in Section 4 and proved in Section 5. Its proof uses the following symmetry. Instances come with an involution σ, the half-turn of the conflict square in Figure 1(a). Explicitly, given $\mathcal{I} = (X_1, X_2, X_3, X_4, \alpha, \beta, R, C)$, its dual instance is

$$\sigma(\mathcal{I}) = (X_3, X_4, X_1, X_2, \beta, \alpha, C^{\mathsf{T}}, R^{\mathsf{T}}),$$

where $C^{\mathsf{T}} = \{(s, i) : (i, s) \in C\}$ and $R^{\mathsf{T}} = \{(j, p) : (p, j) \in R\}$. Transposing coordinates identifies the conflicts, Z-points, and H-points of $\sigma(\mathcal{I})$ with those of $\mathcal{I}$, while exchanging the roles of R and C. In particular, $\Gamma_{HR}(\sigma(\mathcal{I})) \cong \Gamma_{HC}(\mathcal{I})$, $\Gamma_{HC}(\sigma(\mathcal{I})) \cong \Gamma_{HR}(\mathcal{I})$, and $\Gamma_{RZ}(\sigma(\mathcal{I})) \cong \Gamma_{CZ}(\mathcal{I})$.

4. The extremal argument

By Lemmas 2.1 and 2.2 (with $m = n + 1$), Theorem 1.2 implies Theorem 1.1. The argument below is the extremal argument of [15]; we include it to keep the paper self-contained, and to show exactly which form of the deletion lemma is needed.

Proof of Theorem 1.2. Suppose, for a contradiction, that G is a nonempty oriented graph on $X = V(G)$ with minimum indegree at least one in which every nonempty proper subset $U \subsetneq X$ satisfies the strict concavity $|U| + |F^2(U)| < 2|F(U)|$, where $F = F_G$. Because every vertex has an in-neighbour, we have $F(X) = X$.

We apply the concavity to many subsets at once by working with subsets of $X \times X$. For any relation $\Omega \subseteq X \times X$, we define its column-wise and row-wise images under F by

$$F_1\Omega = \{(i,j) : \ (p,j) \in \Omega \text{ and } p \to i\} \qquad \text{and} \qquad F_2\Omega = \{(i,j) : \ (i,s) \in \Omega \text{ and } s \to j\}.$$

In other words, F_1 applies F independently to each column $\Omega_{*j} = \{p : (p,j) \in \Omega\}$, whereas F_2 applies F to each row $\Omega_{i*} = \{s : (i,s) \in \Omega\}$. We call a pair of relations (P,Q) with $P, Q \subseteq X \times X$ *compatible* if their one-step images are disjoint, that is, $F_1P \cap F_2Q = \varnothing$. Let $\Delta = \{(v,v) : v \in X\}$ be the diagonal. The pair (Δ, Δ) is compatible, as $(i,j) \in F_1\Delta \cap F_2\Delta$ would require both $j \to i$ and $i \to j$, impossible in an oriented graph. Among all compatible pairs (P,Q) satisfying $\Delta \subseteq P \cap Q$, we choose one that maximizes the potential function

$$\Phi(P,Q) = |P| + |Q| + d, \qquad \text{where} \quad d = |X|^2 - |F_1P| - |F_2Q|.$$

By compatibility, $d = |X^2 \setminus (F_1P \cup F_2Q)|$ counts the points of $X \times X$ lying outside both one-step images.

We first verify that every column of P and every row of Q is a nonempty proper subset of X. Since $\Delta \subseteq P \cap Q$, each column P_{*j} contains j and each row Q_{i*} contains i. If $P_{*j} = X$ for some j, then $X \times \{j\} = F(X) \times \{j\} \subseteq F_1P$; picking any in-neighbour $w \to j$ in G would then yield $(w,j) \in F_2\Delta \subseteq F_2Q$, contradicting $F_1P \cap F_2Q = \varnothing$. By symmetry, $Q_{i*} \subsetneq X$ for every i. We may therefore apply the strict concavity (3) to each column P_{*j} and each row Q_{i*}. Summing over all $j \in X$ and $i \in X$ gives

$$|P| + |F_1^2P| < 2|F_1P| \qquad \text{and} \qquad |Q| + |F_2^2Q| < 2|F_2Q|.$$

Set $R = X^2 \setminus F_2^2Q$ and $C = X^2 \setminus F_1^2P$. The sum of these two inequalities is then a lower bound for $|R| + |C|$:

$$|R| + |C| = 2|X|^2 - |F_2^2Q| - |F_1^2P| > |P| + |Q| + 2d. \tag{5}$$

These complements are the natural candidates for a better pair: besides (5), membership in R and C is what keeps the H-points of their conflicts away from $F_1P \cup F_2Q$, as we show below. The pair (R,C) need not be compatible, however. Take $X_1 = X_2 = X_3 = X_4 = X$ and let $\alpha = \beta$ be the arc relation of G. Then (R,C) is an instance in the sense of Section 3, and by definition $F_1R \cap F_2C$ is its set Z of Z-points. More generally, for $R' \subseteq R$ and $C' \subseteq C$, a point of Z lies in $F_1R' \cup F_2C'$ if and only if some member of R' or C' covers it. Theorem 3.1 provides conflict-free subfamilies $R' \subseteq R$ and $C' \subseteq C$ that retain all conflict-free members of R and C and satisfy

$$|R \setminus R'| + |C \setminus C'| \le |H| + q, \tag{6}$$

where q is the number of points in Z covered by no member of R' or C'.

First, $\Delta \subseteq R \cap C$, and no diagonal member is in a conflict, so $\Delta \subseteq R' \cap C'$. Indeed, if $(v,v) \in F_2^2Q$, then $(v,s) \in F_2Q$ for some $s \to v$; but $(s,s) \in P$ gives $(v,s) \in F_1P$, contradicting compatibility. Likewise, a conflict between $(v,v) \in R$ and $(i,s) \in C$ would give $s \to v \to i$, and then $(s,s) \in P$ puts (i,s) in F_1^2P, contradicting $(i,s) \in C$. Symmetrically, $\Delta \cap F_1^2P = \varnothing$ and no $(v,v) \in C$ is in a conflict.

Second, $|H| \le d$. Consider any conflict between $(p,j) \in R$ and $(i,s) \in C$, which produces the H-point (p,s) with $p \to i$ and $s \to j$. If (p,s) belonged to F_1P, there would exist some $(u,s) \in P$ with $u \to p$; walking two steps along $u \to p \to i$ would then place (i,s) in F_1^2P, contradicting $(i,s) \in C$. By the same token, if $(p,s) \in F_2Q$, there would exist $(p,t) \in Q$ with $t \to s$, and the two-step walk $t \to s \to j$ would place (p,j) in F_2^2Q, contradicting $(p,j) \in R$. Consequently, H is disjoint from $F_1P \cup F_2Q$, and so $|H| \le d$.

Because R' and C' have no conflicts, we have $F_1R' \cap F_2C' = \varnothing$, so (R', C') is a compatible pair with $\Delta \subseteq R' \cap C'$ and deletion cost $|R \setminus R'| + |C \setminus C'| \le |H| + q \le d + q$. Let $d' = |X|^2 - |F_1R'| - |F_2C'|$ be the number of points outside $F_1R' \cup F_2C'$. As noted above, the q uncovered points of Z all lie outside $F_1R' \cup F_2C'$, ensuring that $q \le d'$. Combining this with (5), we arrive at

$$\Phi(R', C') = |R'| + |C'| + d' \ge |R| + |C| - (d + q) + d' \ge |R| + |C| - d > |P| + |Q| + d = \Phi(P, Q),$$

which contradicts the extremal choice of (P, Q). □

5. Proof of the deletion lemma

The proof is by induction on $|R| + |C|$. Members in no conflict, and the case in which H cannot be matched into one of the families, are handled by short reductions (Section 5.1). What remains is the case in which H can be matched into both R and C. There we delete everything, and the whole difficulty is the following inequality, proved in Section 5.3.

5.1. **Reduction to the counting lemma.**

Lemma 5.1 (Counting lemma)**.** *Suppose that every member of R and C participates in a conflict, and that both Γ_{HR} and Γ_{HC} possess matchings saturating H. Then $|R| + |C| \le |Z| + |H|$.*

Proof of Theorem 3.1, assuming Lemma 5.1. We proceed by induction on $|R| + |C|$, calling any pair (R', C') satisfying the conclusion of the theorem a *solution*. If there are no conflicts, then $\nu(\Gamma_{HR}) = \nu(\Gamma_{HC}) = 0$ and (R, C) itself is a solution. Assume henceforth that at least one conflict exists, and set $\nu^* = \min\{\nu(\Gamma_{HR}), \nu(\Gamma_{HC})\}$.

Removing a member that is in no conflict changes none of Z, H, ν^*, and such a member covers no point of Z (Section 3); so adding it back to a solution of the smaller instance, which exists by induction, gives a solution.

We may therefore assume that every member of R and C participates in at least one conflict. Suppose next that $\nu^* < |H|$. By applying the involution σ if necessary (which exchanges R and C while preserving $|R| + |C|$, ν^*, and q), we may assume without loss of generality that $\nu(\Gamma_{HR}) = \nu^* < |H|$. By the defect form of Hall's theorem, there exists a nonempty subset $H' \subseteq H$ whose neighbourhood $R_{H'} = \bigcup_{h \in H'} R(h)$ in Γ_{HR} satisfies

$$|R_{H'}| \le |H'| - |H| + \nu^*.$$

Because each block $R(h)$ is nonempty, so is $R_{H'}$. We may thus apply the induction hypothesis to the instance $(R \setminus R_{H'}, C)$, whose sets of Z-points, H-points, and uncovered Z-points we denote by Z°, H°, and q°. This gives conflict-free subfamilies $R' \subseteq R \setminus R_{H'}$ and $C' \subseteq C$ satisfying

$$|(R \setminus R_{H'}) \setminus R'| + |C \setminus C'| \le |H^\circ| + q^\circ.$$

(Since every member of the original instance now lies in a conflict, the requirement that (R', C') retain the conflict-free members of the original instance is vacuous, here and in the final case below.) Plainly $Z^\circ \subseteq Z$ and $H^\circ \subseteq H$. No $h \in H'$ can survive in H°, because every conflict with H-point h involves a member of $R(h) \subseteq R_{H'}$, all of which have been deleted; hence $|H^\circ| \le |H| - |H'|$. Moreover, any point of Z° left uncovered by R' and C' is also a point of Z left uncovered by R' and C', so $q^\circ \le q$. Combining these estimates gives

$$|R \setminus R'| + |C \setminus C'| = |R_{H'}| + |(R \setminus R_{H'}) \setminus R'| + |C \setminus C'| \le |R_{H'}| + |H| - |H'| + q \le \nu^* + q,$$

so (R', C') is a solution.

Finally, suppose that every member participates in a conflict and $\nu^* = |H|$, so that both Γ_{HR} and Γ_{HC} admit matchings saturating H. In this case we delete everything by setting $R' = C' = \varnothing$. Then all points of Z are left uncovered ($q = |Z|$), and Lemma 5.1 guarantees that the total deletion cost is $|R| + |C| \le |H| + |Z| = \nu^* + q$. □

5.2. **Greedy bases in transversal matroids.** For subsets $S \subseteq X_3$ and $Q \subseteq X_4$, let $\lambda(S,Q)$ denote the maximum size of a matching in the bipartite subgraph of β induced between S and Q, and write $N_\beta(S) = \{j \in X_4 : s \to j \text{ for some } s \in S\}$. By Kőnig's theorem (or Hall's marriage theorem [10]), this matching number is given by the min-cut formula

$$\lambda(S,Q) = \min_{M \subseteq S} (|S \setminus M| + |N_\beta(M) \cap Q|).$$

If M_1, M_2 attain the minimum for (S_1,Q_1) and (S_2,Q_2), then $M_1 \cup M_2$ and $M_1 \cap M_2$ are admissible for $(S_1 \cup S_2, Q_1 \cap Q_2)$ and $(S_1 \cap S_2, Q_1 \cup Q_2)$; since $N_\beta(M_1 \cap M_2) \subseteq N_\beta(M_1) \cap N_\beta(M_2)$, this shows that λ is *bisubmodular*: for all $S_1, S_2 \subseteq X_3$ and $Q_1, Q_2 \subseteq X_4$,

$$\lambda(S_1,Q_1) + \lambda(S_2,Q_2) \;\geq\; \lambda(S_1 \cup S_2,\, Q_1 \cap Q_2) + \lambda(S_1 \cap S_2,\, Q_1 \cup Q_2). \tag{7}$$

For any $S \subseteq X_3$, $T \subseteq X_4$, and $j \in X_4 \setminus T$, the marginal matching increment

$$\partial_j(S,T) = \lambda(S, T \cup \{j\}) - \lambda(S,T) \in \{0,1\}$$

satisfies, by (7) (with $S_1 = S_2 = S$, $Q_1 = T \cup \{j\}$, $Q_2 = T'$, respectively $(S_1,Q_1) = (S,T)$, $(S_2,Q_2) = (S', T \cup \{j\})$),

$$\partial_j(S,T) \geq \partial_j(S,T') \quad (T \subseteq T' \subseteq X_4 \setminus \{j\}) \qquad \text{and} \qquad \partial_j(S',T) \geq \partial_j(S,T) \quad (S \subseteq S' \subseteq X_3). \tag{8}$$

We call the first inequality *diminishing returns*. Also $\lambda(S,T) = \lambda(S, T \cap N_\beta(S))$, since vertices outside $N_\beta(S)$ are isolated. For fixed S, $Q \mapsto \lambda(S,Q)$ is the rank function of the transversal matroid on X_4 determined by β and S.

Fix once and for all a global linear order $\prec$ on X_4, and for any $Q \subseteq X_4$ and $j \in X_4$ write $Q_{\succ j} = \{x \in Q : j \prec x\}$. The *greedy set* of Q with respect to S, that is, the basis of Q in this matroid obtained by scanning Q in decreasing $\prec$-order, is

$$g(S,Q) = \{j \in Q :\; \partial_j(S, Q_{\succ j}) = 1\}.$$

Lemma 5.2 (Greedy bases)**.** *For every $S \subseteq X_3$ and $Q \subseteq X_4$, we have $|g(S,Q)| = \lambda(S,Q)$, and $g(S,Q)$ can be completely matched into S (that is, $\lambda(S,Q_0) = |Q_0|$ for every $Q_0 \subseteq g(S,Q)$).*

Proof. Telescoping the marginal increments along $\prec$ gives $\lambda(S,Q) = \sum_{j \in Q} \partial_j(S, Q_{\succ j}) = |g(S,Q)|$. Writing $J = g(S,Q)$, we have $J_{\succ j} \subseteq Q_{\succ j}$ for each $j \in J$, so diminishing returns (8) gives $\partial_j(S, J_{\succ j}) \geq \partial_j(S, Q_{\succ j}) = 1$. Summing over $j \in J$ yields $\lambda(S,J) = \sum_{j \in J} \partial_j(S, J_{\succ j}) = |J|$, so J, and hence every $Q_0 \subseteq J$, can be completely matched into S. □

5.3. **Proof of the counting lemma.** The counting lemma combines two inequalities. The first shows, via Hall's theorem and double counting, that when H can be matched into C, the family R can be matched into Z.

Lemma 5.3 (Local Hall lemma)**.** *If every member of R participates in at least one conflict and Γ_{HC} contains a matching saturating H, then Γ_{RZ} contains a matching saturating R.*

Proof. For fixed j, we weight each member $r \in U$ and each point $z \in T$ (both defined below) by the number of vertices s with $s \to j$ through which it can take part in a conflict. The matching saturating H shows that T carries at least as much total weight as U, while passing from a member to an adjacent point cannot increase the weight; for a minimal Hall violator these two facts, together with the positive weight of every member, are incompatible.

The graph Γ_{RZ} decomposes over $j \in X_4$ as the disjoint union of the bipartite graphs Γ^j_{RZ} between the j-fibres of R and Z. Fix $j \in X_4$ and write N_j for neighbourhoods in Γ^j_{RZ}. By Hall's marriage theorem, it suffices to show that $|N_j(U)| \geq |U|$ for every subset U of members $(p,j) \in R$. Suppose, for a contradiction, that some subset violates Hall's condition, and choose a violator U of minimum cardinality, with $T = N_j(U)$. By minimality, for every $w \in U$ the subset $U \setminus \{w\}$ satisfies Hall's condition and thus admits a saturating matching into T. This forces $|T| = |U| - 1$ and provides, for each $w \in U$, a bijection $\mu_w \colon U \setminus \{w\} \to T$ with $\mu_w(r) \in N_j(r)$ for all $r \in U \setminus \{w\}$.

Let $S_j = \{s \in X_3 : s \to j\}$. For each $s \in S_j$, define

$$U_s = \{(p,j) \in U : \ p \to i \text{ for some } (i,s) \in C\} \qquad \text{and} \qquad T_s = \{(i,j) \in T : \ (i,s) \in C\}.$$

If $(p,j) \in U_s$ and $(i,s) \in C$ with $p \to i$, then (p,j) and (i,s) conflict (since $s \to j$), yielding $(p,s) \in H$ and $(i,j) \in T_s$. A matching in Γ_{HC} saturating H maps the distinct H-points $\{(p,s) : (p,j) \in U_s\}$ injectively to members $(i,s) \in C$ with $p \to i$, and hence to distinct points $(i,j) \in T_s$. Thus $|U_s| \le |T_s|$ for every $s \in S_j$. Summing this inequality over all $s \in S_j$ and interchanging the order of summation gives

$$\sum_{r \in U} a_r \le \sum_{z \in T} b_z, \qquad \text{where} \quad a_r = |\{s \in S_j : r \in U_s\}|, \quad b_z = |\{s \in S_j : z \in T_s\}|. \tag{9}$$

On the other hand, whenever $r = (p,j) \in U$ is adjacent to $z = (i,j) \in T$ in Γ^j_{RZ}, we have $p \to i$, which implies that every $s \in S_j$ satisfying $z \in T_s$ also satisfies $r \in U_s$; consequently, $a_r \ge b_z$. Fixing any $w \in U$ and pairing $U \setminus \{w\}$ with T via the bijection μ_w, we find

$$\sum_{r \in U} a_r = a_w + \sum_{r \in U \setminus \{w\}} a_r \ge a_w + \sum_{r \in U \setminus \{w\}} b_{\mu_w(r)} = a_w + \sum_{z \in T} b_z.$$

Finally, since every member of R participates in a conflict, $w = (p,j)$ must conflict with some $(i,s) \in C$, which ensures $s \in S_j$ and $w \in U_s$, so $a_w \ge 1$. This contradicts (9). □

The second inequality, Lemma 5.4 below, comes from linear algebra. Assign an independent indeterminate x_{pi} to each arc $p \to i$ and y_{sj} to each arc $s \to j$, and work over the rational function field $\mathbb{K}$ generated by these indeterminates. Consider the four incidence matrices $L \in \mathbb{K}^{Z \times R}$, $N \in \mathbb{K}^{C \times Z}$, $B \in \mathbb{K}^{H \times R}$, and $A \in \mathbb{K}^{C \times H}$ defined by setting $L_{(i,j),(p,j)} = x_{pi}$ and $A_{(i,s),(p,s)} = x_{pi}$ whenever $p \to i$, setting $N_{(i,s),(i,j)} = y_{sj}$ and $B_{(p,s),(p,j)} = y_{sj}$ whenever $s \to j$, and setting all other entries to 0. A direct check shows that both matrix products NL and AB have $((i,s),(p,j))$-entry equal to $x_{pi}y_{sj}$ if (p,j) and (i,s) conflict, and 0 otherwise; thus the conflict matrix factors through both Z and H as

$$NL = AB.$$

The matrices L, N, and B are block-diagonal (decomposing over $j \in X_4$, $i \in X_2$, and $p \in X_1$, respectively) with distinct indeterminates appearing in each block. By a theorem of Edmonds [5], their generic ranks over $\mathbb{K}$ coincide with the maximum matching numbers $\nu(\Gamma_{RZ})$, $\nu(\Gamma_{CZ})$, and $\nu(\Gamma_{HR})$. Applying Sylvester's rank inequality to $NL = AB$ therefore yields

$$\nu(\Gamma_{RZ}) + \nu(\Gamma_{CZ}) - |Z| = \operatorname{rank} L + \operatorname{rank} N - |Z| \le \operatorname{rank}(NL) = \operatorname{rank}(AB) \le \operatorname{rank} B = \nu(\Gamma_{HR}).$$

We now prove this inequality without matrices.

Lemma 5.4 (Combinatorial Sylvester inequality)**.** *For every instance $\mathcal{I}$, we have*

$$\nu(\Gamma_{RZ}) + \nu(\Gamma_{CZ}) \le |Z| + \nu(\Gamma_{HR}) \le |Z| + |H|.$$

Proof. For each $i \in X_2$, define the slices $C_i = \{s \in X_3 : (i,s) \in C\}$ and $Z^i = \{j \in X_4 : (i,j) \in Z\}$; similarly, for each $p \in X_1$, set $R_p = \{j \in X_4 : (p,j) \in R\}$ and $C^{(p)} = \bigcup_{i : p \to i} C_i$. First, if $p \to i$, then $R_p \cap N_\beta(C_i) \subseteq Z^i$, because any $j \in R_p$ and $s \in C_i$ with $s \to j$ produce a conflict between $(p,j) \in R$ and $(i,s) \in C$ with Z-point (i,j). Second, Γ_{CZ} decomposes as the disjoint union over $i \in X_2$ of the bipartite graphs induced by β between C_i and Z^i, so that $\nu(\Gamma_{CZ}) = \sum_{i \in X_2} \lambda(C_i, Z^i)$.

For each $i \in X_2$, let $J_i = g(C_i, Z^i) \subseteq Z^i$ be the greedy set along $\prec$, and form the combined set $Y = \{(i,j) \in Z : j \in J_i\}$. By Lemma 5.2, we have $|Y| = \sum_i |J_i| = \nu(\Gamma_{CZ})$. Let W denote the set of members $(p,j) \in R$ that cover at least one point $(i,j) \in Y$ (meaning $p \to i$ and $j \in J_i$).

Every member $(p,j) \in R \setminus W$ is adjacent in Γ_{RZ} exclusively to points of $Z \setminus Y$; thus any matching in Γ_{RZ} uses at most $|Z \setminus Y| = |Z| - |Y|$ edges incident to $R \setminus W$ and at most $|W|$ edges incident to W, giving

$$\nu(\Gamma_{RZ}) \le |W| + |Z| - |Y| = |W| + |Z| - \nu(\Gamma_{CZ}).$$

Since $\nu(\Gamma_{HR}) \le |H|$ holds trivially, it remains only to show that W can be completely matched into H in Γ_{HR}, which will give $|W| \le \nu(\Gamma_{HR})$.

The difficulty is that a member $(p, j) \in W$ may cover points of Y in several rows i with $p \to i$, and matchings chosen independently in different rows could send two members (p, j) and (p, j') to the same $s \in X_3$, that is, to the same H-point (p, s). Choosing every J_i greedily along one fixed order prevents this: an element that is greedy for C_i stays greedy for the union $C^{(p)}$ of the rows met by p, so for each p the relevant elements are independent in a single transversal matroid and can be matched at once.

To construct such a matching, fix $p \in X_1$ and consider the slice $W_p = \{j \in X_4 : (p, j) \in W\} \subseteq R_p$. For any $j \in W_p$, we can choose $i \in X_2$ with $p \to i$ and $j \in J_i$. Setting $T = (R_p)_{\succ j}$, we have $T \cap N_\beta(C_i) \subseteq Z^i_{\succ j}$ since $R_p \cap N_\beta(C_i) \subseteq Z^i$. Moreover, $j \in J_i \subseteq Z^i$ implies $j \in N_\beta(C_i)$. Using diminishing returns, the isolation of vertices outside $N_\beta(C_i)$, and the second inequality in (8) for $C_i \subseteq C^{(p)}$, we obtain the chain of inequalities

$$1 = \partial_j(C_i, Z^i_{\succ j}) \le \partial_j(C_i, T \cap N_\beta(C_i)) = \partial_j(C_i, T) \le \partial_j(C^{(p)}, T) \le 1.$$

Hence $\partial_j(C^{(p)}, (R_p)_{\succ j}) = 1$, which means that $W_p \subseteq g(C^{(p)}, R_p)$. Lemma 5.2 therefore supplies an injection $\phi_p \colon W_p \to C^{(p)}$ such that $\phi_p(j) \to j$ for every $j \in W_p$. By definition of $C^{(p)}$, for each $j \in W_p$ the image $s = \phi_p(j)$ lies in $C_{i'}$ for some i' with $p \to i'$; the resulting conflict between $(p, j) \in R$ and $(i', s) \in C$ shows that $(p, s) \in H$ and $(p, j) \in R(p, s)$. Assembling the assignments $(p, j) \mapsto (p, \phi_p(j))$ across all $p \in X_1$ and $j \in W_p$ produces a matching in Γ_{HR} saturating W, completing the proof. □

Proof of Lemma 5.1. Lemma 5.3 applied directly to $\mathcal{I}$ gives $\nu(\Gamma_{RZ}) = |R|$. Applying Lemma 5.3 to the dual instance $\sigma(\mathcal{I})$ (where $\Gamma_{HC}(\sigma(\mathcal{I})) \cong \Gamma_{HR}(\mathcal{I})$ likewise has a matching saturating its H-points) yields $\nu(\Gamma_{CZ}(\mathcal{I})) = \nu(\Gamma_{RZ}(\sigma(\mathcal{I}))) = |C|$. The desired bound now follows from Lemma 5.4. □

Remark 5.5. Theorem 3.1 also yields a sharpening of the auxiliary matching bound in [15]. Let Γ be the bipartite graph with left part $R \sqcup Z_L$ and right part $C \sqcup Z_R$ (where Z_L, Z_R are disjoint copies of Z), with edges joining conflicting pairs $(r, c) \in R \times C$, as well as $r \in R$ to $z_R \in Z_R$ and $z_L \in Z_L$ to $c \in C$ whenever r or c covers $z \in Z$. Given a solution (R', C') from Theorem 3.1, form a vertex set K in Γ by taking all deleted members $(R \setminus R') \cup (C \setminus C')$, the copies z_R of points $z \in Z$ covered by R', and the copies z_L of points $z \in Z$ covered by C'. Since R' and C' are conflict-free, no $z \in Z$ is covered by both R' and C', so $|K| = |R \setminus R'| + |C \setminus C'| + |Z| - q \le |Z| + \min\{\nu(\Gamma_{HR}), \nu(\Gamma_{HC})\}$. By construction, K is a vertex cover of Γ, which implies $\nu(\Gamma) \le |K| \le |Z| + \min\{\nu(\Gamma_{HR}), \nu(\Gamma_{HC})\}$, refining the bound $\nu(\Gamma) \le |Z| + |H|$ of [15].

6. Consequences

We conclude with two consequences of Theorem 1.1. The second is well known to follow from the conjecture (Remark 6.3); we include the short deductions for completeness. See also Seacrest [16] for a related arc-weighted formulation.

Corollary 6.1 (Vertex-weighted form)**.** *For every nonempty oriented graph D and every weight function $w \colon V(D) \to [0, \infty)$, there exists a vertex $v \in V(D)$ such that $w(N_1^+(v)) \le w(N_2^+(v))$, where $w(S) = \sum_{x \in S} w(x)$.*

Proof. Suppose first that w takes positive integer values. Construct a blow-up oriented graph D_w by replacing each vertex $u \in V(D)$ with an independent set $I_u = \{(u, k) : 1 \le k \le w(u)\}$ of size $w(u)$, and directing an arc from (u, k) to (y, l) whenever $u \to y$ in D. Because D has no digons, for any vertex $x = (v, k)$ in D_w we have $N_1^+(x) = \bigcup_{u \in N_1^+(v)} I_u$ and $N_2^+(x) = \bigcup_{u \in N_2^+(v)} I_u$. Applying Theorem 1.1 to D_w therefore yields a vertex $v \in V(D)$ with $w(N_1^+(v)) \le w(N_2^+(v))$. By scaling,

the same holds for all positive rational weight functions, and a standard compactness (closedness) argument extends the conclusion to all $w \in [0,\infty)^{V(D)}$. □

Corollary 6.2 (Directed triangles)**.** *Every oriented graph D on $n \geq 1$ vertices with no directed triangle contains a vertex v satisfying $2|N_1^+(v)|+|N_1^-(v)| \leq n-1$, where $N_1^-(v) = \{u \in V(D) : u \to v\}$. Consequently, any oriented graph on $n \geq 1$ vertices with minimum outdegree and minimum indegree at least $n/3$ contains a directed triangle.*

Proof. By Theorem 1.1, D contains a good vertex v. Since D is oriented and triangle-free, $N_2^+(v)$ cannot intersect $N_1^-(v)$ (as any $w \in N_2^+(v) \cap N_1^-(v)$ would close a two-step walk $v \to u \to w$ into a directed triangle $v \to u \to w \to v$). Thus the four sets $\{v\}$, $N_1^+(v)$, $N_1^-(v)$, and $N_2^+(v)$ are pairwise disjoint, which gives

$$|N_1^+(v)| \leq |N_2^+(v)| \leq n - 1 - |N_1^+(v)| - |N_1^-(v)|.$$

Rearranging gives $2|N_1^+(v)| + |N_1^-(v)| \leq n - 1$, which is violated whenever all in- and out-degrees are at least $n/3$. □

Remark 6.3. The Caccetta–Häggkvist conjecture [2] asserts in particular that every digraph on n vertices with minimum outdegree at least $n/3$ contains a directed cycle of length at most three; for oriented graphs, a directed triangle. Seymour observed that his conjecture implies the weaker version in which both the minimum outdegree and the minimum indegree are at least $n/3$ (see [3, 18]); this is the second statement of Corollary 6.2. This semidegree version was listed as open in Sullivan's survey [18]. The outdegree version remains open to our knowledge; the best bound known is due to Hladký, Král', and Norin [12], who showed that minimum outdegree at least $0.3465n$ forces a directed triangle.

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University of British Columbia, Vancouver, BC, Canada
Email address: qhxie@student.ubc.ca